

A mm-Wave Phase Interpolator Based on Frequency Doubling

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Abstract—This paper presents the design and analysis of a millimeter-wave (mm-wave) phase interpolator architecture based on frequency doubling. By operating the core interpolation circuitry at half the output frequency, the design relaxes speed requirements on the interpolating cells while maintaining fine phase resolution at mm-wave frequencies.

Index Terms—phase interpolator, mm-wave, frequency doubling, phase-locked loop, clock generation, 5G

I. INTRODUCTION

Phase interpolators (PIs) are a critical building block in high-speed serial links and mm-wave clock generation circuits, providing fine-grained phase selection between a set of reference phases. At millimeter-wave frequencies (30 GHz to 300 GHz), the design of PIs is challenged by the reduced transconductance efficiency of transistors, increased sensitivity to parasitic capacitances, and tight timing margins.

A common technique to ease these constraints is frequency doubling: the phase interpolation is performed at a sub-harmonic (typically $f_0/2$) and the output is then frequency-doubled to the target mm-wave frequency. This relaxes the bandwidth requirement on the interpolating core and allows the use of lower-speed, more linear combining topologies.

The remainder of this paper is organized as follows. Section II describes the top-level architecture. Section III details the transistor-level implementation. Section IV provides a theoretical analysis of phase resolution and nonlinearity. Section V presents simulation results, and Section VI concludes the paper.

II. ARCHITECTURE

The top-level system is shown in Fig. 1. A 35 GHz reference is delivered via wirebond, buffered, and split into two quadrature paths by a reflection-based 90° coupler. The direct (in-phase) output of the coupler provides the x -phase input to the current-mode phase interpolator (CMPI), while the coupler output passes through a transmission line to introduce the orthogonal y -phase. The CMPI produces an interpolated phase ϕ_{PI} at 35 GHz, which is then doubled to 70 GHz by a $\times 2$ frequency multiplier. To suppress free-running noise beyond the locking bandwidth, a direct 35 GHz injection path locks the doubler to the reference.

III. CIRCUIT IMPLEMENTATION

A. VGA and Phase Shifter

The variable-gain amplifier (VGA) is implemented as an active-feedback stage with a doublet zero-pole pair that shapes the frequency response across the 30 GHz to 40 GHz target band. An all-pass filter (APF) network provides continuous phase tuning at 35 GHz without disturbing the gain profile, as shown in Fig. 2.

B. Current-Mode Phase Interpolator

IV. THEORETICAL ANALYSIS

A. Phase Interpolation Mechanics

The CMPI synthesises an arbitrary phase ϕ_{interp} between two boundary phasors $\vec{V}_1$ and $\vec{V}_2$ by weighting their tail currents:

$$\vec{V}_{\text{out}} = W_1 \vec{V}_1 + W_2 \vec{V}_2, \quad (1)$$

where W_1 and W_2 are normalised weights summing to unity. Fig. 3 illustrates the construction for $\vec{V}_1$ at 120° and $\vec{V}_2$ at 150°, producing a resultant at 140°.

B. Phase Noise Through the Frequency Doubler

For a generic $\times N$ frequency multiplier, phase noise transforms as

$$\mathcal{L}_{\text{out}}(f_\Delta) = \mathcal{L}_{\text{in}}(f_\Delta) + 20 \log_{10}(N) + \mathcal{L}_{\text{add}}(f_\Delta), \quad (2)$$

where $\mathcal{L}_{\text{add}}$ is the additive noise contribution of the doubler itself. For $N = 2$ the irreducible degradation is

$$\Delta\mathcal{L} = 20 \log_{10}(2) \approx 6 \text{ dB}. \quad (3)$$

C. Absolute Jitter Preservation

Time jitter σ_t relates to integrated phase variance σ_ϕ^2 and carrier frequency f_c by

$$\sigma_t = \frac{\sigma_\phi}{2\pi f_c}. \quad (4)$$

After the $\times 2$ doubler the phase variance scales as $N^2 \sigma_\phi^2$ while the frequency becomes $N f_c$, yielding

$$\sigma_{t,\text{out}} = \frac{N \sigma_{\phi,\text{in}}}{2\pi(N f_{\text{in}})} = \sigma_{t,\text{in}}. \quad (5)$$

Equation (5) shows that an ideal frequency doubler preserves absolute jitter: the 6 dB phase noise degradation is exactly cancelled by the factor-of-two increase in carrier frequency.

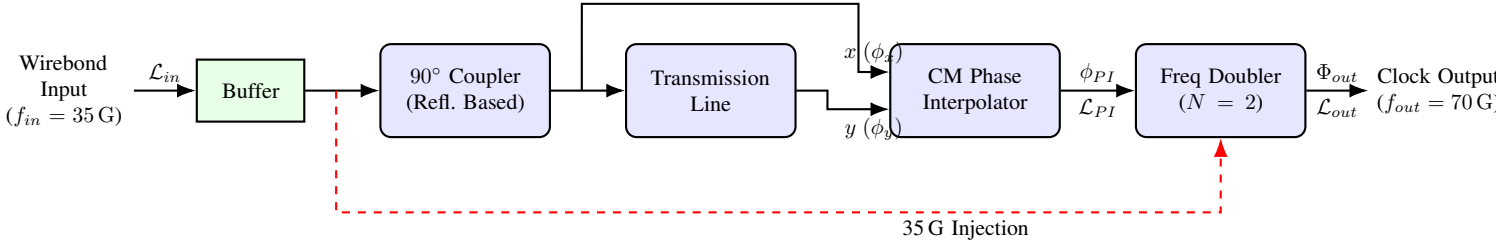


Fig. 1. Top-level 70 GHz clock generation architecture. The CMPI operates at 35 GHz; a $\times 2$ doubler produces the mm-wave output. Direct injection locking suppresses doubler free-running noise.

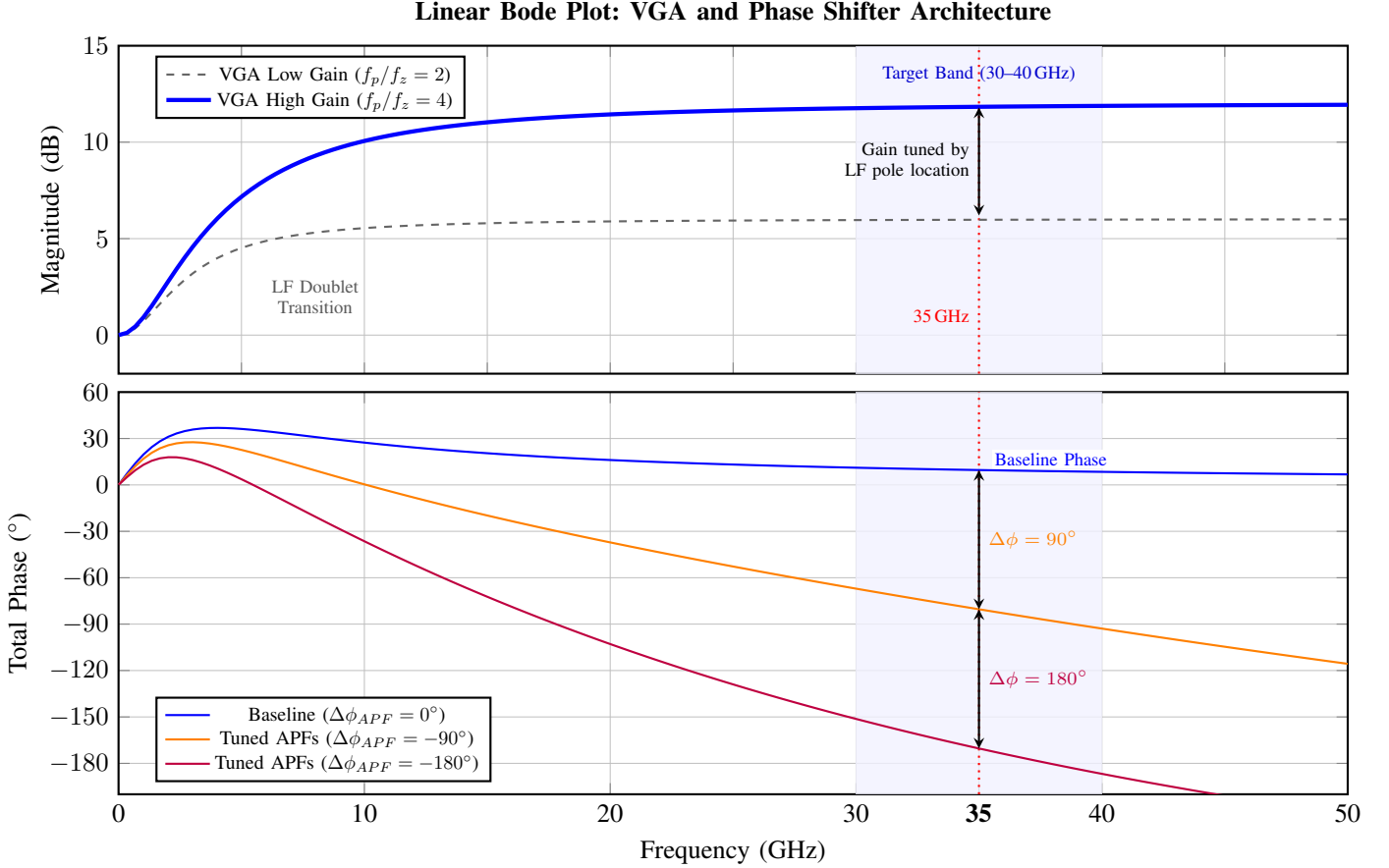


Fig. 2. Bode plot of the VGA and all-pass phase shifter network. Top: gain magnitude across the 30 GHz to 40 GHz band for two pole-zero ratios. Bottom: total phase at 35 GHz is tunable over 180° by adjusting the APF pole frequency.

D. Injection Locking Dynamics

Beyond the free-running doubler noise floor, injection locking confines the output phase noise to the $\mathcal{L}_{in} + 6$ dB bound within the locking bandwidth f_L . Outside f_L the output reverts to the free-running profile, as illustrated in Fig. 4.

V. SIMULATION RESULTS

Fig. 4 shows the conceptual phase noise profiles at key nodes. The 35 GHz input trace follows a combined flicker-plus-white noise model. The ideal doubled output sits 6 dB above the input in accordance with (3). The free-running 70 GHz doubler exhibits a noisier profile with a 50 MHz noise corner; injection

locking pulls it to the ideal +6 dB bound within the locking bandwidth f_L , beyond which it tracks its free-running noise floor.

VI. CONCLUSION

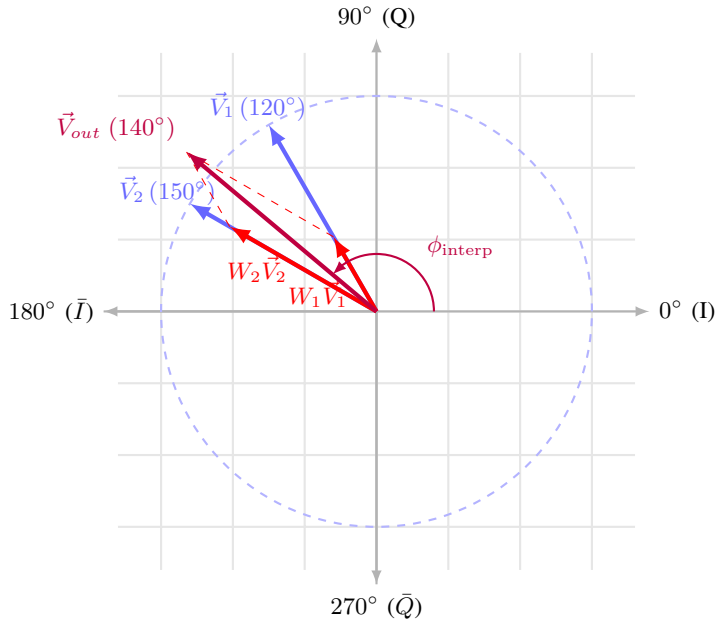


Fig. 3. Phasor construction for the CMPI. Blue: boundary phasors $\vec{V}_1$ (120°) and $\vec{V}_2$ (150°). Red: weighted components $W_1\vec{V}_1$, $W_2\vec{V}_2$. Purple: resultant interpolated phase at 140°. The weight ratio $W_2/W_1 = \sin(\phi - \phi_1)/\sin(\phi_2 - \phi)$ follows from the sine rule.

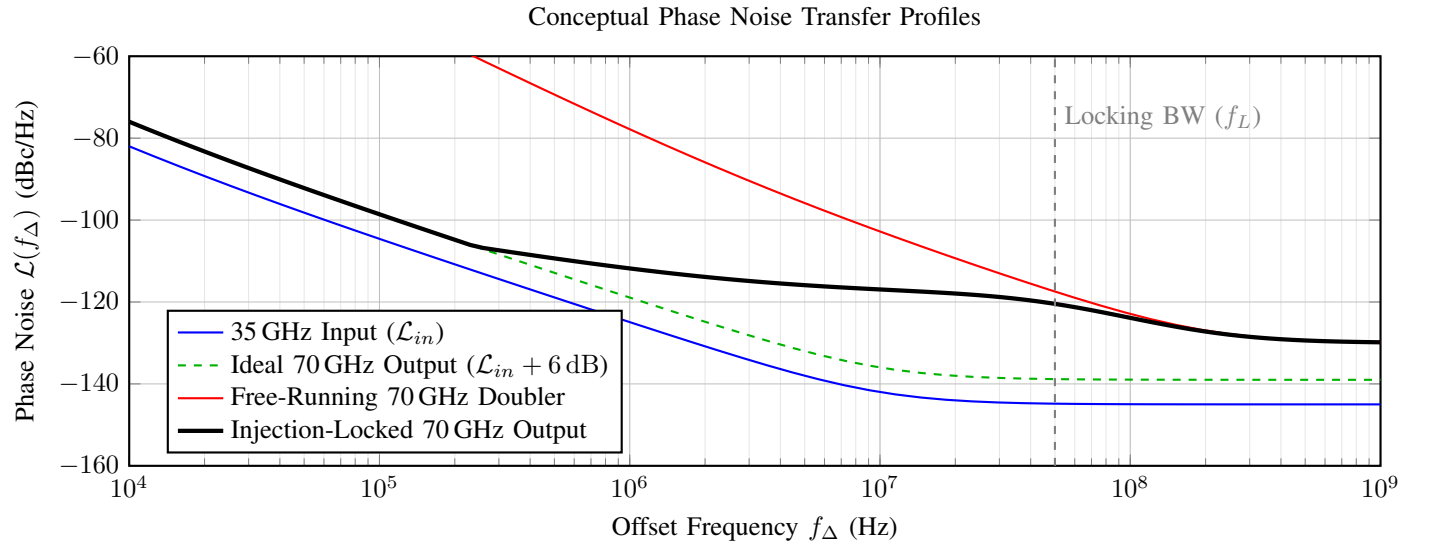


Fig. 4. Conceptual phase noise at the 35 GHz input, the ideal doubled output (+6 dB), the free-running doubler, and the injection-locked output. Within the locking bandwidth $f_L \approx 50$ MHz the output tracks the ideal curve; beyond f_L it reverts to the free-running profile.